Maximum length of a 3-sum-free set consisting of 2 intervals

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June 24, 2026

Theorem 1. *Let A be a subset of $\mathbb{T}$ consisting of 2 disjoint closed intervals with measure at least $\frac{1}{5}$. Then, $A + A - 3 \cdot A = \mathbb{T}$.*

Proof. By monotonicity, it is enough to prove the result in the case $\mu(A) = \frac{1}{5}$. Indeed, if $\mu(A) > \frac{1}{5}$, then A contains a subset A', still consisting of 2 disjoint closed intervals, with measure exactly $\frac{1}{5}$. If $A' + A' - 3 \cdot A' = \mathbb{T}$, then, since $A' \subset A$, also $A+A-3\cdot A = \mathbb{T}$. Thus we may assume from now on that $\mu(A) = \frac{1}{5}$. We may also assume that both intervals have positive length. Indeed, if one of them has length 0, then the other has length $\frac{1}{5}$, and for an interval K of length $\frac{1}{5}$ we have $K + K - 3K = \mathbb{T}$.

Note that if the statement is true for a set A, then it is also true for any translate of A and for the set $-A$. Without loss of generality, we can therefore assume $A = [0, \alpha] \cup [x, x + \beta]$ satisfying the following conditions:

$$0 < \alpha < x < x + \beta < 1, \tag{1}$$

$$\alpha \geq \beta, \tag{2}$$

$$x - \alpha \leq 1 - (x + \beta), \qquad \text{i.e.} \qquad x \leq \frac{1 + \alpha - \beta}{2}. \tag{3}$$

Indeed, we first choose an interval of maximal length, call its length α, call the other length β, and translate so that the longer interval starts at 0. We can then write

$$A = [0, \alpha] \cup [x, x + \beta], \qquad 0 < \alpha < x < x + \beta < 1, \qquad \alpha \geq \beta.$$

Let

$$g_1 = x - \alpha, \qquad g_2 = 1 - (x + \beta)$$

be the two gaps between the intervals. If $g_1 \leq g_2$, then (3) already holds. If $g_1 > g_2$, then we replace A by the reflected and translated set $\alpha - A$. This keeps the longer interval anchored at 0, since

$$\alpha - A = [0, \alpha] \cup [1 + \alpha - x - \beta, 1 + \alpha - x].$$

Writing $x' = 1 + \alpha - x - \beta$, the new gaps are

$$x' - \alpha = 1 - (x + \beta) = g_2, \qquad 1 - (x' + \beta) = x - \alpha = g_1.$$

Thus the two gaps have been exchanged, and since in this case $g_2 < g_1$, the transformed set satisfies

$$x' - \alpha \leq 1 - (x' + \beta).$$

Intuitively, the reflection reverses which side of the longer interval the larger gap lies on; the final translation simply puts that longer interval back in the form $[0, \alpha]$. Furthermore, since $\mu(A) = \frac{1}{5}$, we have $\alpha + \beta = \frac{1}{5}$, and therefore by (2), $\alpha \geq \frac{1}{10}$ and $\beta \leq \frac{1}{10}$. Let us denote $I = [0, \alpha]$ and $J = [x, x + \beta]$. We will prove that $\mathbb{T}$ is covered by the following 5 intervals:

$$\begin{aligned}
R_1 &= I + I - 3 \cdot I = [-3\alpha, 2\alpha], \\
R_2 &= I + J - 3 \cdot I = [-3\alpha + x, \alpha + x + \beta], \\
R_3 &= J + J - 3 \cdot I = [2x - 3\alpha, 2x + 2\beta], \\
R_4 &= I + I - 3 \cdot J = [-3x - 3\beta, -3x + 2\alpha], \\
R_5 &= I + J - 3 \cdot J = [-2x - 3\beta, \alpha + \beta - 2x].
\end{aligned}$$

Note that all of these intervals as written are of the form $[a, b]$ where $0 < b - a < 1$, (we say that the intervals are in *proper* form) and therefore checking if a point $y \in \mathbb{T}$ is contained in one of these intervals is equivalent to checking if for some integer k, $y + k \in [a, b]$.

We denote $U = R_1 \cup R_2 \cup R_3 \cup R_4 \cup R_5$ and $V = \mathbb{T} \setminus U$. We assume for the sake of contradiction that V is non-empty. Identifying $\mathbb{T}$ with $[0, 1)$, note that $V \subset \mathbb{T} \setminus R_1 = (2\alpha, 1 - 3\alpha)$, which again is in proper form. The rest of the proof consists in iteratively shrinking this interval through a series of claims on the intervals R_i until we show that V is empty. The first claim is the following:

Claim 1. $[2\alpha, \alpha + x + \beta] \subset R_2$.

Proof. We just need to show that $2\alpha \in R_2$. In fact, we prove that

$$-3\alpha + x \leq 2\alpha \leq \alpha + x + \beta. \tag{4}$$

The first inequality is equivalent to $x \leq 5\alpha$. By assumption (3), we have

$$x \leq \frac{1 + \alpha - \beta}{2} \leq \frac{1 + \alpha - (1/5 - \alpha)}{2} = \alpha + \frac{2}{5}.$$

and since $\alpha \geq \frac{1}{10}$, we have $\alpha + \frac{2}{5} \leq 5\alpha$. The second inequality is equivalent to $\alpha \leq x + \beta$, which is true by assumption (1). $\square$

We can then conclude that

$$V \subset (\alpha + x + \beta, 1 - 3\alpha). \tag{5}$$

Furthermore, if $\alpha + x + \beta \geq 1 - 3\alpha$, then the previous claim implies that R_2 covers the whole interval $(2\alpha, 1 - 3\alpha)$, and hence V is empty, contradicting our assumption. Therefore

$$1 - 3\alpha > \alpha + x + \beta \qquad \text{i.e.} \qquad x < 1 - 4\alpha - \beta. \tag{6}$$

In particular, the interval in (5) is in proper form. We also note that from inequality (6) we also get the weaker but more convenient

$$x < 1 - (\alpha + \beta) - 3\alpha \leq 1 - 1/5 - 3/10 = 1/2. \tag{7}$$

Claim 2. *$[\alpha + \beta + x, 2x + 2\beta] \subset R_3$.*

Proof. Similarly to the previous claim, it suffices to show that

$$2x - 3\alpha \leq \alpha + \beta + x \leq 2x + 2\beta. \tag{8}$$

The first inequality is equivalent to $x \leq 4\alpha + \beta$. From condition (3) and the fact that $\alpha + \beta = \frac{1}{5}$, we have

$$x \leq \frac{1 + \alpha - \beta}{2} = \frac{\alpha - \beta + 5(\alpha + \beta)}{2} = 3\alpha + 2\beta. \tag{9}$$

But since $\alpha \geq \beta$, we have $3\alpha + 2\beta \leq 4\alpha + \beta$, and we are done.

The second inequality is equivalent to $\alpha \leq x + \beta$, which is true by assumption (1). □

The conclusion of this is that in fact

$$V \subset (2x + 2\beta, 1 - 3\alpha). \tag{10}$$

Similarly as before, if $2x + 2\beta \geq 1 - 3\alpha$, then the previous claim implies that R_3 covers the whole interval $(\alpha + x + \beta, 1 - 3\alpha)$, and hence V is empty, contradicting our assumption. Therefore

$$1 - 3\alpha > 2x + 2\beta \qquad \text{i.e.} \qquad x < \frac{1 - 3\alpha - 2\beta}{2}. \tag{11}$$

and that the interval in (10) is in proper form. Again, we can obtain a weaker but more convenient inequality from (11):

$$x < \frac{1 - 3\alpha - 2\beta}{2} \leq \frac{1 - 2/5 - 1/10}{2} = 1/4. \tag{12}$$

The proof will be completed once we show that R_4 and R_5 cover the interval $(2x + 2\beta, 1 - 3\alpha)$. We first show that the union of these two intervals is itself an interval:

Claim 3. *The ends of the intervals R_4 and R_5 satisfy the following inequalities:*

$$-3x - 3\beta \leq -2x - 3\beta \leq -3x + 2\alpha \leq \alpha + \beta - 2x.$$

and therefore

$$R_4 \cup R_5 = [-3x - 3\beta, \alpha + \beta - 2x]. \tag{13}$$

Furthermore, this interval is in proper form.

Proof. The first inequality is equivalent to $x \geq 0$, which is true by assumption (1). The second inequality is equivalent to $x \leq 2\alpha + 3\beta$. Now, since $x \leq 1/4$ by (12), and since $2\alpha + 3\beta = 2/5 + \beta \geq 2/5 > 1/4$, this inequality is also true. The third inequality is equivalent to $\alpha \leq x + \beta$, which is true by assumption (1). To see that it is in proper form, compute its length directly:

$$(\alpha + \beta - 2x) - (-3x - 3\beta) = x + \alpha + 4\beta.$$

Using (12) and $\alpha + \beta = \frac{1}{5}$, we get

$$x + \alpha + 4\beta < \frac{1}{4} + \alpha + 4\beta = \frac{1}{4} + \frac{1}{5} + 3\beta \leq \frac{1}{4} + \frac{1}{5} + \frac{3}{10} = \frac{3}{4} < 1,$$

as required. $\square$

It remains only to show that the interval in (13) covers the interval in (10):

Claim 4. $R_4 \cup R_5 \supset (2x + 2\beta, 1 - 3\alpha)$.

Proof. We have calculated the ends of both intervals in proper form. Therefore, it suffices to show that

$$-3x - 3\beta \leq (2x + 2\beta) - 1 \leq (1 - 3\alpha) - 1 \leq \alpha + \beta - 2x,$$

where we have shifted both points to check by the integer -1. We have already established the middle inequality. The first inequality is equivalent to $5(x + \beta) \geq 1$, which is true because $x + \beta \geq \alpha + \beta = 1/5$. The last inequality is equivalent to $2x \leq 4\alpha + \beta = 3\alpha + 1/5$. Since $x \leq 1/4$ by (12), and we have $\alpha \geq 1/10$, we get

$$2x \leq 1/2 = 3/10 + 1/5 \leq 3\alpha + 1/5,$$

and we are done. $\square$

We have therefore shown that V is empty, so the proof is complete.

$\square$